Topic 5: Solved Problems

Step-by-step solutions of problems 1 to 4 of the 15 of the official sheet on mechanical properties (Problems T5_MechanicalProperties: the tensile test, the elastic modulus, yield and tensile strength, ductility, resilience, true stress and strain, thermal stress and hardness, from Sessions 11 and 12), written in four phases. The sheet has no worked solutions, only a numeric key for most items, so every solution is derived here and compared with that key.

`Problem 1` `Problems T5_MechanicalProperties.pdf #1` `Tensile test · full stress-strain analysis`

Complete Analysis of a Tensile Test from the Load-Elongation Table

`Original statement · Problems T5_MechanicalProperties.pdf`

1. A cylindrical aluminum sample has a diameter of 12.8 mm and an initial base length of 12.5 mm and it is subjected to uniaxial tension. Use the data from the load/elongation table to answer the following questions:

- *a. Represent the engineering stress data versus the engineering strain.*
- *b. Calculate Young's modulus.*
- *c. Calculate the yield strength, assuming that the residual strain at the yield point is 0.002.*
- *d. Work out the tensile strength of this alloy.*
- *e. Determine the approximate value of ductility expressed as a deformation percentage.*
- *f. Determine the approximate value of the modulus of resilience.*
- *g. Calculate the load (N) and the elongation (mm) of the sample at the yield point.*
- *h. Calculate an approximate working stress for this material.*
- *i. With the values obtained, is it possible to calculate the load applied to the sample if the elongation produced is 0.1 mm? If it is possible, work it out and if it is not possible, explain your answer and provide an alternative method.*

Point	F (N/100)	ΔL (mm)
1	127	0.006
2	254	0.012
3	381	0.019
4	508	0.025
5	762	0.038
6	891	0.051
7	927	0.063
8	1025	0.095
9	1078	0.127
10	1194	0.190

Point	F (N/100)	ΔL (mm)
11	1283	0.254
12	1497	0.508
13	1590	0.762
14	1604	0.889
15	1595	1.016
16	1515	1.270
17	1247	1.460

(The sheet prints the table in two rows, F and ΔL, followed by the word "Fracture"; here each point has its own row. Point 17 is the last record before the fracture.)

Official Solution: b. $E \approx 200$ **GPa; c.** $\sigma_y = 740$ **MPa; d.** $\sigma_{max} = 1250$ **MPa; e. 11.3 %; f.** $U_r = 1.37$ **MPa; g.** $F = 9.5 \times 10^4$ **N,** $\Delta l = 0.071$ **mm; h.** $\sigma_w = 370$ **MPa; i. it can only be obtained graphically,** $F = 10.3 \times 10^4$ **N**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** The load acts along the axis of the cylinder, and up to the necking the stress is the same on every cross-section of the gauge length. Under this hypothesis the engineering stress and strain are defined with the original area and length [Session 11 Slide 9].
2. **Linear elastic start.** For small loads the curve is a straight line through the origin, Hooke's law $\sigma = E\varepsilon$, and the slope of that line is the Young's modulus [Session 11 Slide 13].
3. **Conventional yield strength.** The yield strength is the stress at which the curve crosses the line drawn parallel to the elastic line from the residual strain 0.002 (0.2 % offset), as the statement asks [Session 11 Slide 28].
4. **Between table rows the curve is a straight segment.** The table gives discrete points, so every reading between two of them is a linear interpolation.
5. **Elastic recovery.** When the sample breaks the load disappears and the elastic strain is recovered, so the ductility is the plastic strain at fracture [Session 11 Slide 32].
6. **Safety factor.** The statement gives none; the slide says that N is usually 2, and that value is used for the working stress [Session 11 Slide 29].

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	12.8 mm	12.8×10^{-3} m
Initial gauge length	l_0	12.5 mm	12.5×10^{-3} m
Load (table)	F	$100 \times (F/100)$	N
Elongation (table)	ΔL	see table	mm

Quantity	Symbol	Value	In SI
Safety factor	N	2	

Unknowns: the curve (a), E (b), σ_y (c), σ_{max} (d), the ductility (e), U_r (f), F_y and Δl_y (g), σ_w (h), and the load at $\Delta L = 0.1$ mm (i).

PHASE 2 · FRAMES & STRATEGY

There is a single direction, the axis of the sample, and the stress state is uniaxial, so no change of basis and no rotation matrix are involved.

Strategy. Every item comes from the same table of engineering stress and strain, so the first thing to do is to build it:

$$\sigma_n = \frac{F}{A_0} \qquad \varepsilon_n = \frac{\Delta L}{l_0}$$

- The Young's modulus is the slope of the straight initial part (Hooke's law). A straight line through the origin is fitted by least squares: it minimises $\sum(\sigma_i - E\varepsilon_i)^2$, which gives $E = \sum \sigma_i\varepsilon_i / \sum \varepsilon_i^2$.
- The yield strength is found by the 0.2 % offset construction, solved algebraically on the segment of the table where the curve crosses the offset line.
- The tensile strength is the maximum nominal stress, and since A_0 is constant it corresponds to the maximum load.
- The ductility is the plastic strain at fracture, and the modulus of resilience is the area under the elastic line up to the yield point, $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35].
- For the load at a given elongation the table is used again: Hooke's law only holds in the elastic zone.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(12.8\ \text{mm})^2}{4}$$

The square is $12.8^2 = 163.84$, then $\pi \times 163.84 = 514.72$ and dividing by 4:

$$A_0 = 128.68\ \text{mm}^2 = 1.2868 \times 10^{-4}\ \text{m}^2$$

2. Engineering stress and strain of every point

The load of the table is in hundreds of newtons, so $F = 100 \times (F/100)$. For point 1:

$$F_1 = 100 \times 127 = 12\,700\ \text{N} \qquad \sigma_{n,1} = \frac{12\,700\ \text{N}}{128.68\ \text{mm}^2} = 98.69\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{0.006\ \text{mm}}{12.5\ \text{mm}} = 4.80 \times 10^{-4}$$

The same operation for all the points (the last column is the secant modulus, used in item b):

Point	F (N)	ΔL (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
1	12 700	0.006	98.69	0.00048	205.6
2	25 400	0.012	197.39	0.00096	205.6
3	38 100	0.019	296.08	0.00152	194.8
4	50 800	0.025	394.78	0.00200	197.4
5	76 200	0.038	592.17	0.00304	194.8
6	89 100	0.051	692.42	0.00408	169.7
7	92 700	0.063	720.39	0.00504	142.9
8	102 500	0.095	796.55	0.00760	104.8
9	107 800	0.127	837.74	0.01016	82.5
10	119 400	0.190	927.89	0.01520	61.0
11	128 300	0.254	997.05	0.02032	49.1
12	149 700	0.508	1163.35	0.04064	28.6
13	159 000	0.762	1235.63	0.06096	20.3
14	160 400	0.889	1246.51	0.07112	17.5
15	159 500	1.016	1239.51	0.08128	15.2
16	151 500	1.270	1177.34	0.10160	11.6
17	124 700	1.460	969.07	0.11680	8.3

a) Stress-strain curve

The points of the table are plotted as they are, joined by straight segments. The right-hand panel enlarges the elastic part, where the Young's modulus line and the offset line of item c are drawn.

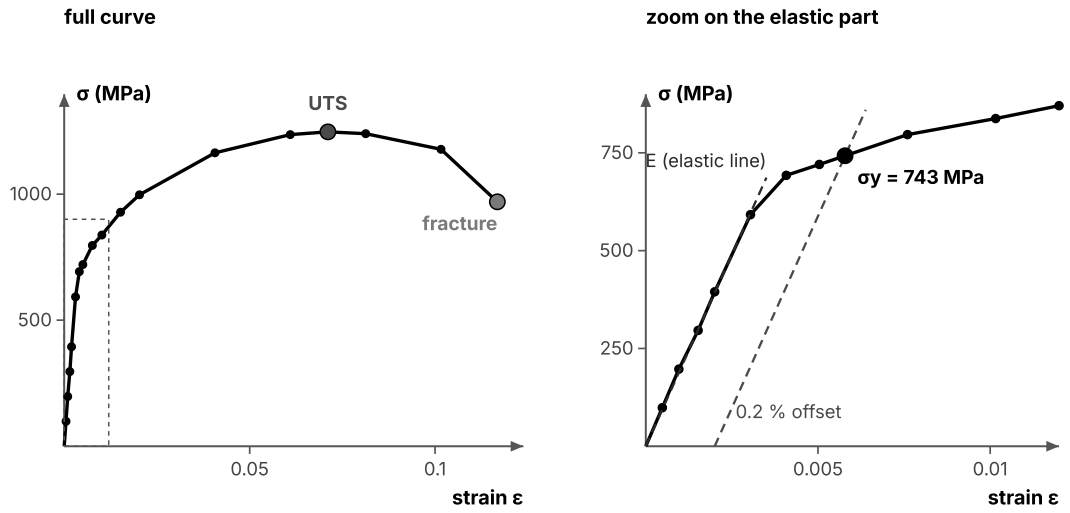


Figure 1. Problem 1: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Young's modulus

The secant modulus σ_n/ε_n of the last column is almost constant, 195 to 206 GPa, for points 1 to 5, and it drops to 170 GPa at point 6 and to 143 GPa at point 7: the specimen leaves the linear zone after point 5. The straight part is therefore made of points 1 to 5, and the fit through the origin is:

Point	ε_i	σ_i (MPa)	$\sigma_i\varepsilon_i$ (MPa)	ε_i^2
1	0.00048	98.69	0.0474	2.304×10^{-7}
2	0.00096	197.39	0.1895	9.216×10^{-7}
3	0.00152	296.08	0.4500	2.310×10^{-6}
4	0.00200	394.78	0.7896	4.000×10^{-6}
5	0.00304	592.17	1.8002	9.242×10^{-6}

$$\sum \sigma_i\varepsilon_i = 3.277 \text{ MPa} \qquad \sum \varepsilon_i^2 = 1.6704 \times 10^{-5}$$

$$E = \frac{\sum \sigma_i\varepsilon_i}{\sum \varepsilon_i^2} = \frac{3.277}{1.6704 \times 10^{-5}} = 196160 \text{ MPa} = \mathbf{196.2\ GPa} \approx 2.0 \times 10^{2} \text{ GPa}$$

c) Yield strength (0.2 % offset)

The offset line starts at $\varepsilon = 0.002$ and is parallel to the elastic line:

$$\sigma_{off}(\varepsilon) = E\,(\varepsilon - 0.002)$$

The curve is above the line while $g = \sigma_n - E(\varepsilon_n - 0.002)$ is positive. The sign of g changes between points 7 and 8, so the crossing is on that segment:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
6	0.00408	692.4	408.0	284.4
7	0.00504	720.4	596.3	124.1
8	0.00760	796.6	1098.5	-301.9
9	0.01016	837.7	1600.7	-762.9

The segment between the two points is the line

$$\sigma(\varepsilon) = \sigma_7 + m\,(\varepsilon - \varepsilon_7) \qquad m = \frac{\sigma_8 - \sigma_7}{\varepsilon_8 - \varepsilon_7} = \frac{796.55 - 720.39}{0.00760 - 0.00504} = 29749.3\ \text{MPa}$$

Equating it to the offset line, $\sigma_7 + m(\varepsilon - \varepsilon_7) = E(\varepsilon - 0.002)$, and solving for ε:

$$\varepsilon^{*} = \frac{\sigma_7 - m\,\varepsilon_7 + 0.002\,E}{E - m} = \frac{720.39 - 29749.3 \times 0.00504 + 0.002 \times 196160}{196160 - 29749.3} = \frac{962.778}{166411.2} = 0.0057855$$

$$\sigma_y = E\,(\varepsilon^{*} - 0.002) = 196160 \times (0.0057855 - 0.002) = \mathbf{742.6\ MPa}$$

d) Tensile strength

The maximum load of the table is $F_{max} = 160\,400$ N at point 14. The nominal stress uses the original area, so the maximum load gives the maximum stress:

$$\sigma_{max} = \frac{F_{max}}{A_0} = \frac{160\,400\ \text{N}}{128.68\ \text{mm}^2} = \mathbf{1246.5\ MPa}$$

e) Ductility

The last record before the fracture is point 17: $\Delta L = 1.460$ mm and $F = 124\,700$ N, so

$$\varepsilon_f = \frac{1.460}{12.5} = 0.1168 \qquad \sigma_f = \frac{124\,700}{128.68} = 969.1 \text{ MPa}$$

At that moment the strain is the sum of an elastic and a plastic part, $\varepsilon_t = \varepsilon_e + \varepsilon_p$. After the fracture the load is zero and the elastic part is recovered along a line parallel to the elastic one, so what remains is ε_p [Session 11 Slide 32]:

$$\varepsilon_e = \frac{\sigma_f}{E} = \frac{969.1}{196160} = 0.00494 \qquad \varepsilon_p = \varepsilon_f - \varepsilon_e = 0.1168 - 0.00494 = 0.1119$$

$$EL\,(\%) = \varepsilon_p \times 100 = \mathbf{11.2\ \%}$$

f) Modulus of resilience

The area under the elastic line up to the yield point is a triangle, $U_r = \tfrac{1}{2}\sigma_y \varepsilon_y$ with $\varepsilon_y = \sigma_y / E$ [Session 11 Slide 35]:

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(742.6 \text{ MPa})^2}{2 \times 196160 \text{ MPa}} = \mathbf{1.41\ MPa}$$

($1 \text{ MPa} = 1 \text{ MJ/m}^3$.)

g) Load and elongation at the yield point

The load is the yield stress times the original area, and the elongation is the total strain of the yield point (the value ε^* of item c) times the original length:

$$F_y = \sigma_y A_0 = 742.6 \text{ MPa} \times 128.68 \text{ mm}^2 = 95\,554 \text{ N} = \mathbf{95.6\ kN}$$

$$\Delta l_y = \varepsilon^* l_0 = 0.0057855 \times 12.5 \text{ mm} = \mathbf{0.0723\ mm}$$

h) Working stress

$$\sigma_w = \frac{\sigma_y}{N} = \frac{742.6}{2} = \mathbf{371.3\ MPa}$$

i) Load for an elongation of 0.1 mm

The elongation of the yield point is 0.0723 mm, so 0.1 mm is in the plastic zone and the load is not proportional to the elongation. Using Hooke's law anyway, $F = EA_0\Delta l/l_0$, would give

$$F = \frac{196160\ \mathrm{N/mm^2} \times 128.68\ \mathrm{mm^2} \times 0.1\ \mathrm{mm}}{12.5\ \mathrm{mm}} = 201\,935\ \mathrm{N} = 202\ \mathrm{kN}$$

which is larger than the maximum load that the sample supports, 160 400 N: the answer is **not** possible with the values obtained. The alternative is the table itself. The elongation 0.1 mm is between points 8 ($\Delta L = 0.095$ mm, $F = 102\,500$ N) and 9 ($\Delta L = 0.127$ mm, $F = 107\,800$ N), and a linear interpolation on that segment gives

$$F = F_8 + \frac{0.1 - 0.095}{0.127 - 0.095}\,(F_9 - F_8) = 102\,500 + 0.15625 \times (107\,800 - 102\,500) = \mathbf{103\,328\ N}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $\sigma_n = F/A_0$ in N/mm² is MPa, ε_n is a ratio of lengths and has no units, so $E = \sigma/\varepsilon$ is in MPa and $U_r = \sigma_y^2/2E$ is in MPa (energy per unit volume).
- **Order.** $\sigma_y = 742.6$ MPa $< \sigma_{max} = 1246.5$ MPa, and the fracture stress (969.1 MPa) is below the maximum, as in a ductile sample that necks.
- **The fit.** The secant moduli of points 1 to 5 are within 4.8 % of the fitted E, and the offset line crosses the curve between points 7 and 8, inside the segment used.
- **Item i.** The interpolated load, 103.3 kN, is between the loads of the two rows and below the maximum load; the Hooke value, 202 kN, is not.

Interpretation

The sample is stiff ($E \approx 196$ GPa) and strong, and it yields at about 742.6 MPa, a strain of 0.579 %. From there to the maximum load the strain hardening raises the stress by 504 MPa, and the 11.2 % of plastic elongation at fracture shows that it is also ductile.

THE DATA ARE NOT THOSE OF ALUMINIUM

The table of Session 11 gives $E = 69$ GPa for aluminium and 207 GPa for steel [Session 11 Slide 17]. The data of this problem give $E \approx 196$ GPa and a yield strength above 700 MPa, which are the values of a steel, not of aluminium. The statement says "aluminum"; the data are used as they are.

Check against the official solution

The key printed on the sheet is compared with the values computed here:

Item	Official	Computed	Comment
b. E	≈ 200 GPa	196.2 GPa	agrees (2 %)
c. σ_y	740 MPa	742.6 MPa	agrees
d. σ_{max}	1250 MPa	1246.5 MPa	agrees (rounded)
e. ductility	11.3 %	11.2 %	agrees within the reading
f. U_r	1.37 MPa	1.41 MPa	agrees (2.5 %); the key uses $E = 200$ GPa and $\sigma_y = 740$ MPa
g. F_y, Δl_y	9.5×10^4 N, 0.071 mm	95 554 N, 0.0723 mm	agrees
h. σ_w	370 MPa	371.3 MPa	agrees
i. load at 0.1 mm	10.3×10^4 N, graphically	103\,328 N	agrees

READING OF THE TABLE

The text of the PDF prints the first load of the table as "1127", but the figure of the sheet shows "127". With 127 the first point gives $\sigma/\varepsilon = 206$ GPa, in line with the rest of the elastic points and with the key, so 127 (12 700 N) is used.

`Problem 2` `Problems T5_MechanicalProperties.pdf #2` `Tensile test · true stress · area reduction`

Tensile Test of a 15 mm Cylinder: Strength, Resilience, True Stress and Area Reduction

`Original statement · Problems T5_MechanicalProperties.pdf`

1. A cylindrical sample made from a metallic material (assume a Poisson ratio of ν = 0.25) of 15 mm in diameter and a calibrated length of 50 mm was subjected to a tensile test, and the results are shown in the table below:

Point	Force (kN)	ΔL (mm)
1	70	0.25
2	120	0.40
3	150	0.50
4	160	0.60
5	170	0.75
6	200	1.75
7	220	3.00
8	233	5.00
9	233	6.50
10	220	8.00

The diameter of the sample after fracture was 12.45 mm. Calculate:

- *a. The tensile strength.*
- *b. The Young's Modulus (or elastic modulus).*
- *c. The yield stress.*
- *d. The modulus of resilience.*
- *e. The real tension for an engineering deformation of 0.02.*
- *f. The percentage of deformation to fracture and percentage of area reduction.*

(The sheet prints the table in two rows; here each point has its own row.)

Official Solution: a. $\sigma_{UTS} = 1319$ **MPa; b.** $E = 85$ **GPa; c.** $\sigma_y = 920$ **MPa; d.** $U_r = 4.98$ **MPa; e.** $\sigma_r = 1030$ **MPa; f. 15 %, 7.3 %**

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Uniaxial, uniform stress.** Up to the necking the stress is uniform along the gauge length, so the engineering quantities use the original area and length [Session 11 Slide 9].

2. **Linear elastic start.** Hooke's law $\sigma = E\varepsilon$ holds on the first points of the curve [Session 11 Slide 13].
3. **Yield strength by the 0.2 % offset.** The statement does not name a criterion; the conventional one for metals is the 0.2 % offset [Session 11 Slide 28].
4. **Straight segments between table rows**, so readings between two rows are linear interpolations.
5. **Uniform deformation before the necking.** The volume is constant, $A\,l = A_0 l_0$, until the maximum load, where the necking starts [Session 11 Slides 30 and 38]. The true stress of item e is computed in that range.
6. **Elastic recovery after the fracture.** The elastic strain disappears when the load does, and what is left is the plastic strain [Session 11 Slide 32]. The reduction of area uses the measured final diameter [Session 11 Slide 33].
7. **The Poisson ratio is not needed.** $\nu = 0.25$ is a datum of the statement, but none of the six items uses it: the only lateral information is the final diameter, which is measured.

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	15 mm	15×10^{-3} m
Gauge length	l_0	50 mm	50×10^{-3} m
Diameter after fracture	d_f	12.45 mm	12.45×10^{-3} m
Poisson ratio (not used)	ν	0.25	
Load, elongation	F, ΔL	table	kN, mm

Unknowns: σ_{UTS} (a), E (b), σ_y (c), U_r (d), σ_r at $\varepsilon_n = 0.02$ (e), the elongation at fracture and the reduction of area (f).

PHASE 2 · FRAMES & STRATEGY

The only direction is the axis of the cylinder and the stress is uniaxial, so there is no change of basis.

Strategy. As in any tensile test, the first step is to turn the table into engineering stress and strain,

$$\sigma_n = \frac{F}{A_0} \qquad \varepsilon_n = \frac{\Delta L}{l_0}$$

and then read each item from it: the maximum stress (a), the slope of the straight part by least squares through the origin (b), the offset construction solved on the crossing segment (c), $U_r = \sigma_y^2 / 2E$ (d) [Session 11 Slide 35].

True stress (e). With constant volume, $A\,l = A_0 l_0$, so $A = A_0 l_0 / l = A_0/(1+\varepsilon_n)$. The true stress is then

$$\sigma_r = \frac{F}{A} = \frac{F}{A_0}\,(1+\varepsilon_n) = \sigma_n\,(1+\varepsilon_n)$$

which is the relation of the slide for the uniform deformation [Session 11 Slide 38].

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15\ \text{mm})^2}{4}$$

The square is $15^2 = 225$, then $\pi \times 225 = 706.86$ and dividing by 4:

$$A_0 = 176.71\ \text{mm}^2$$

2. Engineering stress and strain of every point

The loads are in kN, so $F = 1000 \times F_{kN}$ newtons. For example, point 1 is $F = 70\,000$ N and $\Delta L = 0.25$ mm, so $\sigma_n = 70\,000/176.71 = 396.1$ MPa and $\varepsilon_n = 0.25/50 = 0.0050$. All the points:

Point	F (N)	ΔL (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
1	70 000	0.25	396.1	0.0050	79.2
2	120 000	0.40	679.1	0.0080	84.9
3	150 000	0.50	848.8	0.0100	84.9
4	160 000	0.60	905.4	0.0120	75.5
5	170 000	0.75	962.0	0.0150	64.1
6	200 000	1.75	1131.8	0.0350	32.3
7	220 000	3.00	1244.9	0.0600	20.7
8	233 000	5.00	1318.5	0.1000	13.2
9	233 000	6.50	1318.5	0.1300	10.1
10	220 000	8.00	1244.9	0.1600	7.8

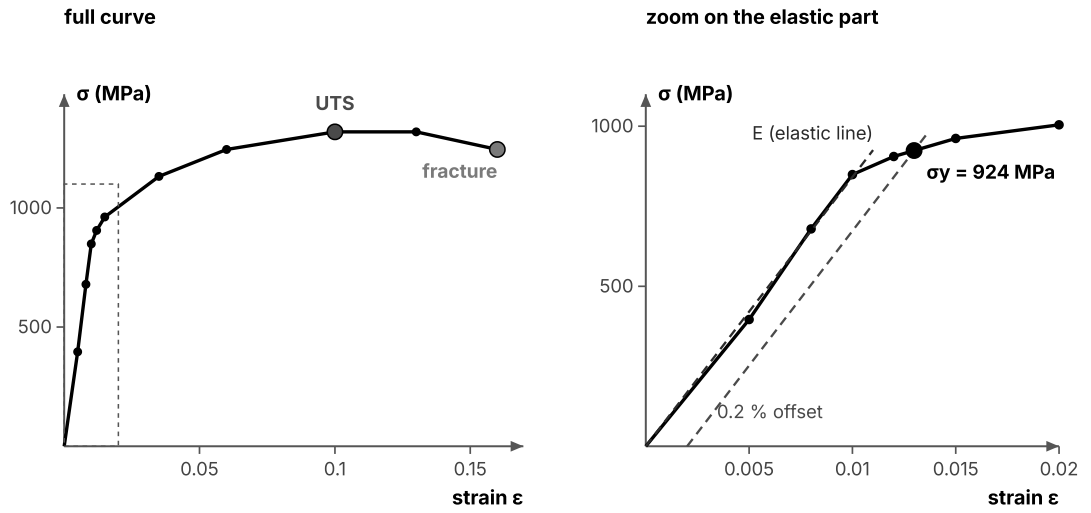


Figure 2. Problem 2: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

a) Tensile strength

The maximum load is $F_{max} = 233\,000$ N, reached at points 8 and 9 ($\varepsilon_n = 0.100$ and 0.130: the load stays at its maximum while the sample elongates). The area A_0 is constant in the engineering stress, so

$$\sigma_{UTS} = \frac{F_{max}}{A_0} = \frac{233\,000\ \text{N}}{176.71\ \text{mm}^2} = \mathbf{1318.5\ MPa}$$

b) Young's modulus

The secant modulus is 79 to 85 GPa for points 1 to 3, and it drops to 75 GPa at point 4 and to 64 GPa at point 5: the linear zone ends after point 3. The fit through the origin with points 1 to 3 is:

Point	ε_i	σ_i (MPa)	$\sigma_i \varepsilon_i$ (MPa)	ε_i^2
1	0.0050	396.1	1.9806	2.500×10^{-5}
2	0.0080	679.1	5.4325	6.400×10^{-5}
3	0.0100	848.8	8.4883	1.000×10^{-4}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{15.901}{1.8900 \times 10^{-4}} = 84134\ \text{MPa} = \mathbf{84.1\ GPa}$$

c) Yield stress (0.2 % offset)

The offset line is $\sigma_{off} = E(\varepsilon - 0.002)$ and the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ tells where the curve crosses it. The sign changes between points 4 and 5:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
3	0.01000	848.8	673.1	175.8
4	0.01200	905.4	841.3	64.1
5	0.01500	962.0	1093.7	-131.7
6	0.03500	1131.8	2776.4	-1644.7

On that segment

$$m = \frac{962.00 - 905.41}{0.01500 - 0.01200} = 18862.8 \text{ MPa} \qquad \varepsilon^* = \frac{\sigma_4 - m\,\varepsilon_4 + 0.002\,E}{E - m} = \frac{905.41 - 18862.8 \times 0.01200 + 0.002 \times 84134}{84134 - 18862.8} = \frac{847.329}{65271.3} = 0.0129817$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 84134 \times (0.0129817 - 0.002) = \mathbf{923.9\ MPa}$$

d) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(923.9\ \text{MPa})^2}{2 \times 84134\ \text{MPa}} = \mathbf{5.07\ MPa}$$

($1\ \text{MPa} = 1\ \text{MJ/m}^3$.)

e) True stress for $\varepsilon_n = 0.02$

The elongation that corresponds to $\varepsilon_n = 0.02$ is $\Delta L = 0.02 \times 50 = 1.00$ mm, which is not a row of the table. It is between points 5 ($\Delta L = 0.75$ mm, $F = 170$ kN) and 6 ($\Delta L = 1.75$ mm, $F = 200$ kN):

$$F = F_5 + \frac{1.00 - 0.75}{1.75 - 0.75}\,(F_6 - F_5) = 170 + \frac{1.00 - 0.75}{1.75 - 0.75}\,(200 - 170) = 177.5\ \text{kN}$$

$$\sigma_n = \frac{177\,500\ \text{N}}{176.71\ \text{mm}^2} = 1004.4\ \text{MPa}$$

This strain is below the one of the maximum load ($\varepsilon_n = 0.100$), so the deformation is still uniform and the relation of Phase 2 applies:

$$\sigma_r = \sigma_n\,(1+\varepsilon_n) = 1004.4 \times (1+0.02) = \mathbf{1024.5\ MPa}$$

f) Elongation at fracture and reduction of area

The last row of the table is the fracture: $\Delta L = 8.00$ mm and $F = 220\,000$ N, so

$$\varepsilon_f = \frac{8.00}{50} = 0.160 \qquad \sigma_f = \frac{220\,000}{176.71} = 1244.9\ \text{MPa}$$

The elastic part of that strain is recovered when the sample breaks, $\varepsilon_e = \sigma_f / E$, and what is left is the plastic strain [Session 11 Slide 32]:

$$\varepsilon_e = \frac{1244.9}{84134} = 0.01480 \qquad \varepsilon_p = 0.160 - 0.01480 = 0.1452 \qquad EL = \mathbf{14.5\ \%}$$

For the reduction of area, the final area from the final diameter is

$$A_f = \frac{\pi\,(12.45\ \text{mm})^2}{4} = \frac{\pi \times 155.0025}{4} = 121.74\ \text{mm}^2$$

$$RA\,(\%) = \frac{A_0 - A_f}{A_0} \times 100 = \frac{176.71 - 121.74}{176.71} \times 100 = \mathbf{31.1\ \%}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** Stresses in N/mm² = MPa, strains without units, E and U_r in MPa.
- **Order.** $\sigma_y = 923.9$ MPa $< \sigma_{UTS} = 1318.5$ MPa, and the stress rises 395 MPa between the yield and the maximum load.

- **True and engineering stress.** The true stress of item e, 1024.5 MPa, is 2 % above the engineering one, 1004.4 MPa, as the factor $1 + \varepsilon_n$ says.
- **Elongation against area reduction.** $RA = 31.1$ % is larger than $EL = 14.5$ %, as expected when a ductile sample necks: the section shrinks much more at the neck than the length grows over the whole gauge length.
- **Neck.** The stress that acts on the neck at the fracture is $F_f / A_f = 220\,000\ \mathrm{N} / 121.74\ \mathrm{mm}^2 = 1807$ MPa, far above the engineering fracture stress of 1244.9 MPa because the area is much smaller than A_0.

Interpretation

The sample reaches 1.3 GPa with a plateau of maximum load, and it fails after a clear neck. The measured final diameter, not Poisson's ratio, gives the lateral information: Poisson's ratio only describes the elastic part of the deformation.

Check against the official solution

Item	Official	Computed	Comment
a. σ_{UTS}	1319 MPa	1318.5 MPa	agrees
b. E	85 GPa	84.1 GPa	agrees (1 %); the key corresponds to the slope between points 2 and 3
c. σ_y	920 MPa	923.9 MPa	agrees (0.5 %)
d. U_r	4.98 MPa	5.07 MPa	agrees (2 %); the key uses $E = 85$ GPa and $\sigma_y = 920$ MPa
e. σ_r	1030 MPa	1024.5 MPa	agrees (0.5 %): the key reads $\sigma_n \approx 1010$ MPa at $\varepsilon_n = 0.02$
f. elongation	15 %	14.5 %	agrees (rounded)
f. area reduction	7.3 %	31.1 %	**does not agree**

DISCREPANCY WITH THE OFFICIAL SOLUTION: AREA REDUCTION

The official 7.3 % cannot be reproduced from the data of the statement. With $d_0 = 15$ mm and $d_f = 12.45$ mm the final area is $A_f = 121.74$ mm² and the reduction is 31.1 %. A reduction of 7.3 % would need a final area of $0.927\ A_0$, that is, a final diameter of $15\sqrt{0.927} = 14.44$ mm. The computed value is kept.

`Problem 3` `Problems T5_MechanicalProperties.pdf #3` `Tensile test · true stress and strain · design`

Tensile Test of a 13 mm Sample: Modulus, Cable Section and True Values at Maximum Load and Fracture

`Original statement · Problems T5_MechanicalProperties.pdf, p. 2`

1. *An uniaxial tensile test was performed on a cylindrical sample of 13 mm in diameter getting the results of load and elongation shown in the table below:*

- *a. Calculate the engineering stress and strain values and draw the stress-strain curve.*
- *b. Calculate the value of the elastic modulus (Young's Modulus) and determine the length of a bar of 13 cm on which a tension of 200 MPa is applied.*
- *c. What is the recommended cross-section for a cable that has to support an elevator with a maximum weight of 450 kg?*
- *d. Calculate the real stress and real strain values at maximum load and at the fracture point, assuming that the diameter of the sample is 12.922 mm and 10.348 mm respectively. Do the calculations without using the engineering stress and strain values.*
- *e. Calculate the modulus of resilience.*
- *f. Calculate the elastic strain at the moment before fracture.*

Point	Load (kg)	Calibrated length (mm)	Engineering stress (MPa)	Engineering strain
0	0	5.148		
1	500	5.151		
2	1500	5.156		
3	2500	5.161		
4	3500	5.166		
5	3750	5.225		
6	3950	5.354		
7	4000	5.457		
8	4000	5.560		
9	3600	5.676		

(In the sheet the table has only the first two columns filled; the last two are left blank to be completed in item a. Point 0 is the unloaded sample, points 7 and 8 have the same, maximum, load, and point 9 is the fracture.)

Official Solution: b. $E = 75.4$ **GPa,** $l = 13.04$ **cm; c.** $A_0 \geq 16.9$ **mm²; d. maximum load:** $\sigma_{UTS} = 300$ **MPa,** $\varepsilon_{UTS} = 0.012$**; fracture:** $\sigma_{fracture} = 420$ **MPa,** $\varepsilon_{fracture} = 0.456$**; e.** $U_r = 0.457$ **MPa; f.** $\varepsilon_y = 3.4 \times 10^{-3}$

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress** up to the necking: engineering stress and strain with the original area and length [Session 11 Slide 9].
2. **Weight to force.** The loads are masses in kilograms; the force is $F = m\,g$ with $g = 9.81$ m/s².
3. **Linear elastic start** [Session 11 Slide 13] and **0.2 % offset yield strength** [Session 11 Slide 28].
4. **Calibrated length.** The first row ($m = 0$) gives the initial length $l_0 = 5.148$ mm and the other rows the current length. The strain is a ratio, so only the proportion matters.
5. **True quantities from the measured diameters** (item d). The true strain is $\varepsilon_r = \ln(A_0/A) = 2\ln(D_0/D)$ and the true stress is $\sigma_r = F/A$ with A the instantaneous area [Session 11 Slide 38]. They are valid also after the necking, because they use the actual area at the neck.
6. **Design stress (item c).** The statement does not give a safety factor. The minimum section is the one for which the cable does not yield, $\sigma = F/A_0 \le \sigma_y$; with the usual factor $N = 2$ it would be larger [Session 11 Slide 29].

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	13 mm	13×10^{-3} m
Initial calibrated length	l_0	5.148 mm	5.148×10^{-3} m
Gravity	g	9.81 m/s²	
Bar of item b: length, stress	l_b, σ	13 cm, 200 MPa	0.13 m, 200×10^6 Pa
Elevator (item c)	m	450 kg	
Diameter at maximum load, at fracture	D_m, D_f	12.922 mm, 10.348 mm	

Unknowns: the curve (a), E and the length of the bar (b), the section of the cable (c), the true stress and strain at maximum load and at fracture (d), U_r (e), the elastic strain before the fracture (f).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis) and uniaxial stress: there is no change of basis.

Strategy. The table is converted into $\sigma_n = F/A_0$ and $\varepsilon_n = (l - l_0)/l_0$. Then:

- E is the slope of the straight part (least squares through the origin), and the bar of item b is a direct application of Hooke's law, $\Delta l = (\sigma/E)\, l_b$.
- σ_y by the 0.2 % offset is needed twice: for the cable (c) and for the resilience (e).
- The true values (d) are computed from the forces and the measured diameters, as the statement asks, and not from the engineering values.
- The elastic strain before the fracture (f) is σ/E with the stress that acts at that moment [Session 11 Slide 32].

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section and conversions

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(13\ \text{mm})^2}{4}$$

With $13^2 = 169$ and $\pi \times 169 = 530.93$:

$$A_0 = 132.73\ \text{mm}^2$$

For the first loaded point, $F_1 = 500\ \text{kg} \times 9.81\ \text{m/s}^2 = 4\,905\ \text{N}$, so

$$\sigma_{n,1} = \frac{4\,905\ \text{N}}{132.73\ \text{mm}^2} = 36.95\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{5.151 - 5.148}{5.148} = 0.00058$$

a) Engineering stress and strain, and the curve

All the points:

Point	Load (kg)	l (mm)	$F = mg$ (N)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
0	0	5.148	0	0.00	0	—
1	500	5.151	4 905	36.95	0.00058	63.4
2	1500	5.156	14 715	110.86	0.00155	71.3

Point	Load (kg)	l (mm)	$F = mg$ (N)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
3	2500	5.161	24 525	184.77	0.00253	73.2
4	3500	5.166	34 335	258.68	0.00350	74.0
5	3750	5.225	36 788	277.16	0.01496	18.5
6	3950	5.354	38 750	291.94	0.04002	7.3
7	4000	5.457	39 240	295.63	0.06002	4.9
8	4000	5.560	39 240	295.63	0.08003	3.7
9	3600	5.676	35 316	266.07	0.10256	2.6

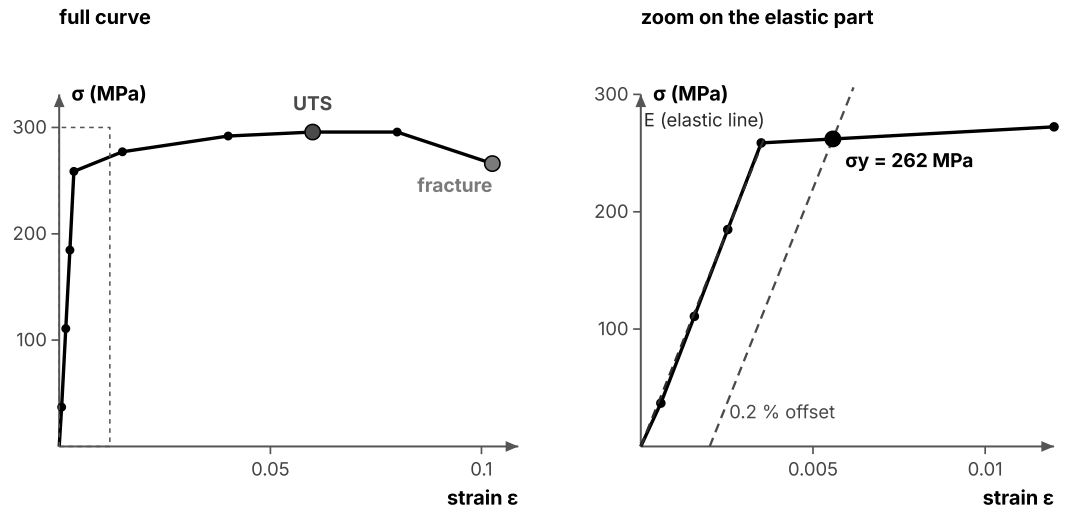


Figure 3. Problem 3: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Young's modulus and the length of the bar

The secant modulus of the last column grows from 63 to 74 GPa in points 1 to 4 and falls to 18 GPa at point 5: the linear zone ends after point 4. The first value is lower because a strain of 0.0006 comes from an elongation of only 0.003 mm, at the limit of the resolution of the table. The fit through the origin on points 1 to 4 is:

Point	ε_i	σ_i (MPa)	$\sigma_i\varepsilon_i$ (MPa)	ε_i^2
1	0.00058	36.95	0.0215	3.396×10^{-7}
2	0.00155	110.86	0.1723	2.415×10^{-6}
3	0.00253	184.77	0.4666	6.377×10^{-6}
4	0.00350	258.68	0.9045	1.223×10^{-5}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{1.565}{2.1357 \times 10^{-5}} = 73273 \text{ MPa} = \mathbf{73.3\ GPa}$$

For the bar, the strain produced by $\sigma = 200$ MPa is $\varepsilon = \sigma / E$ and the elongation is $\Delta l = \varepsilon\, l_b$:

$$\varepsilon = \frac{200}{73273} = 0.002730 \qquad \Delta l = 0.002730 \times 130 \text{ mm} = 0.355 \text{ mm}$$

$$l = l_b + \Delta l = 130 + 0.355 = 130.355 \text{ mm} = \mathbf{13.04\ cm}$$

c) Section of the cable

The yield strength is needed first. With the offset line $E(\varepsilon - 0.002)$ the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ changes between points 4 and 5:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
3	0.00253	184.8	38.5	146.3
4	0.00350	258.7	109.7	149.0
5	0.01496	277.2	949.4	-672.3
6	0.04002	291.9	2785.5	-2493.6

$$m = \frac{277.16 - 258.68}{0.01496 - 0.00350} = 1612.2 \text{ MPa} \qquad \varepsilon^* = \frac{258.68 - 1612.2 \times 0.00350 + 0.002 \times 73273}{73273 - 1612.2} = \frac{399.587}{71660.3} = 0.0055761$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 73273 \times (0.0055761 - 0.002) = \mathbf{262.0\ MPa}$$

The weight of the elevator is $W = 450 \times 9.81 = 4414.5$ N. For the cable not to yield, $W / A_0 \le \sigma_y$:

$$A_0 \geq \frac{W}{\sigma_y} = \frac{4414.5\ \text{N}}{262.0\ \text{N/mm}^2} = \mathbf{16.85\ mm^2}$$

If the usual safety factor $N = 2$ is also applied, the design stress is $\sigma_y/2$ and the section doubles, $A_0 \geq 33.7$ mm².

d) True stress and strain at maximum load and at fracture

Maximum load. $F_m = 4000 \times 9.81 = 39\,240$ N and $D_m = 12.922$ mm:

$$A_m = \frac{\pi\,(12.922)^2}{4} = \frac{\pi \times 166.978}{4} = 131.14\ \text{mm}^2$$

$$\sigma_{r,UTS} = \frac{F_m}{A_m} = \frac{39\,240}{131.14} = \mathbf{299.2\ MPa} \qquad \varepsilon_{r,UTS} = 2\ln\frac{D_0}{D_m} = 2\ln\frac{13}{12.922} = 2\ln(1.00604) = 2 \times 0.006018 = \mathbf{0.01204}$$

Fracture. $F_f = 3600 \times 9.81 = 35\,316$ N and $D_f = 10.348$ mm:

$$A_f = \frac{\pi\,(10.348)^2}{4} = \frac{\pi \times 107.081}{4} = 84.10\ \text{mm}^2$$

$$\sigma_{r,f} = \frac{F_f}{A_f} = \frac{35\,316}{84.10} = \mathbf{419.9\ MPa} \qquad \varepsilon_{r,f} = 2\ln\frac{13}{10.348} = 2\ln(1.25628) = 2 \times 0.22816 = \mathbf{0.4563}$$

e) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(262.0\ \text{MPa})^2}{2 \times 73273\ \text{MPa}} = \mathbf{0.469\ MPa}$$

f) Elastic strain before the fracture

The stress of the table at the fracture point (point 9) is $\sigma_n = 266.1$ MPa, and the elastic part of the strain is given by Hooke's law:

$$\varepsilon_e = \frac{\sigma_n}{E} = \frac{266.1}{73273} = \mathbf{3.63 \times 10^{-3}}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** Masses in kg and g in m/s² give newtons, N/mm² is MPa, the strains have no units, and $A_0 \geq W/\sigma_y$ comes out in mm².
- **Order.** The true stress at maximum load (299.2 MPa) is above the engineering maximum (295.6 MPa) because the area has shrunk, and the true fracture stress (419.9 MPa) is much above the engineering one (266.1 MPa), while the engineering stress falls after the maximum.
- **Limit.** For $D = D_0$ the true strain $2\ln(D_0/D)$ is zero and the true stress is F/A_0, the engineering one.
- **Bar.** The elongation, 0.355 mm over 130 mm, is a strain of 0.27 %, well inside the elastic zone (the yield strain is 3.58×10^{-3} = 0.36 %).

Interpretation

The sample is a ductile metal with a clear yield at 262.0 MPa and a long hardening plateau up to 295.6 MPa; the neck makes the true stress at fracture 419.9 MPa. The cable needs at least 16.85 mm² to stay elastic under the elevator, and about twice as much with the usual safety factor.

> DIAMETERS AND LENGTHS OF THE TABLE DO NOT DESCRIBE THE SAME STRAIN
>
> At the maximum load the length of the table gives $\varepsilon_n = 0.0600$, that is, $\varepsilon_r = \ln(1+\varepsilon_n) = 0.0583$ with the uniform-deformation relation, and constant volume would give a diameter $13/\sqrt{1+\varepsilon_n} = 12.63$ mm. The statement gives 12.922 mm, with $\varepsilon_r = 0.01204$. At the fracture, $A_0/A_f = 1.578$ while $l/l_0 = 1.1026$. The two sets of data do not describe the same deformation: the diameters measure the strain of the neck, the lengths the average over the calibrated length. The statement asks to compute the true values from the diameters, and that is what is done.

Check against the official solution

Item	Official	Computed	Comment
b. E	75.4 GPa	73.3 GPa	agrees (3 %)
b. length of the bar	13.04 cm	13.04 cm	agrees
c. A_0	≥ 16.9 mm²	≥ 16.85 mm²	agrees: the key uses σ_y without safety factor
d. $\sigma_{r,UTS}$, $\varepsilon_{r,UTS}$	300 MPa, 0.012	299.2 MPa, 0.01204	agrees

Item	Official	Computed	Comment
d. $\sigma_{r,f}$, $\varepsilon_{r,f}$	420 MPa, 0.456	419.9 MPa, 0.4563	agrees
e. U_r	0.457 MPa	0.469 MPa	agrees (2.5 %): the key uses $E = 75.4$ GPa
f. elastic strain	3.4×10^{-3}	3.63×10^{-3}	differs 7 %

DISCREPANCY WITH THE OFFICIAL SOLUTION: ELASTIC STRAIN BEFORE THE FRACTURE

The key gives 3.4×10^{-3}. With the stress of the table at the fracture, 266.1 MPa, and the fitted modulus the elastic strain is 3.63×10^{-3}; with the key's modulus, 75.4 GPa, it is 3.53×10^{-3}. The yield strain, $\sigma_y / E = 3.58 \times 10^{-3}$, is also not the key value. The value cannot be reproduced exactly, and the computed one is kept.

Problem 4 | Problems T5_MechanicalProperties.pdf #4 | Tensile test · rectangular brass sample

Tensile Test of a Rectangular Brass Sample: Modulus, Strengths, Resilience and Working Stress

Original statement · Problems T5_MechanicalProperties.pdf, p. 2

1. A tensile test was performed on a brass sample with a rectangular section and dimensions of 3.2 mm x 19.1 mm. Using the data in the table below, answer the following questions:

- *a. Represent the data in terms of engineering stress versus strain.*
- *b. Calculate the modulus of elasticity (Young's modulus).*
- *c. Determine the yield strength.*
- *d. Determine the tensile strength.*
- *e. Calculate the modulus of resilience.*
- *f. Calculate the working stress of this material.*

Point	Load (N)	Length (mm)	σn (MPa)	εn
0	0	63.50		
1	1 380	63.53		
2	2 780	63.58		
3	5 630	63.63		
4	7 430	63.70		
5	8 140	63.75		
6	9 870	64.14		
7	12 850	65.41		
8	14 100	66.68		
9	14 300	67.95		
10	13 830	69.22		
11	12 500	70.49		

(The sheet prints the table in two rows, load and length, with two more rows, σn and εn, left blank to be completed. Point 0 is the unloaded sample.)

Official Solution: **b.** $E \approx 44$ **GPa;** **c.** $\sigma_y = 150$ **MPa;** **d.** $\sigma_{UTS} = 234$ **MPa;** **e.** $U_r = 0.28$ **MPa;** **f.** $\sigma_w = 75$ **MPa**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** The section is a rectangle 3.2×19.1 mm, and up to the necking the stress is uniform: $\sigma_n = F/A_0$ and $\varepsilon_n = (l - l_0)/l_0$ [Session 11 Slide 9].
2. **Initial length.** The first value of the length row, 63.50 mm (load 0), is l_0.
3. **Linear elastic start**, Hooke's law, with the slope fitted by least squares through the origin [Session 11 Slide 13].
4. **0.2 % offset yield strength** [Session 11 Slide 28].
5. **Working stress** $\sigma_w = \sigma_y/N$ with the usual $N = 2$, which the statement does not give [Session 11 Slide 29].
6. **Resolution of the length.** The length is given to 0.01 mm, so the first elongations (0.03 and 0.08 mm) carry an error of 10 to 30 % of their value. This matters for the modulus, and it is analysed in item b.

Data

Quantity	Symbol	Value	In SI
Section	$b \times h$	19.1×3.2 mm	$19.1 \times 10^{-3} \times 3.2 \times 10^{-3}$ m
Initial length	l_0	63.50 mm	63.50×10^{-3} m
Load, length	F, l	table	N, mm
Safety factor	N	2	

Unknowns: the curve (a), E (b), σ_y (c), σ_{UTS} (d), U_r (e), σ_w (f).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the sample), uniaxial stress, no change of basis.

Strategy. The table is turned into σ_n and ε_n. Then E is the least-squares slope through the origin of the straight part, σ_y comes from the offset construction on the segment that crosses the offset line, σ_{UTS} is the maximum load over A_0, $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35] and $\sigma_w = \sigma_y/N$. Since the first points of this table are not perfectly straight, the result is also shown for different choices of points, to see how much it depends on the choice.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = b\,h = 19.1 \text{ mm} \times 3.2 \text{ mm} = 61.12 \text{ mm}^2$$

For point 1: $F_1 = 1\,380$ N and $l_1 = 63.53$ mm, so

$$\sigma_{n,1} = \frac{1\,380\ \text{N}}{61.12\ \text{mm}^2} = 22.58\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{63.53 - 63.50}{63.50} = 0.00047$$

a) Engineering stress and strain, and the curve

All the points:

Point	F (N)	l (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
0	0	63.50	0	0.00000	—
1	1 380	63.53	22.58	0.00047	47.8
2	2 780	63.58	45.48	0.00126	36.1
3	5 630	63.63	92.11	0.00205	45.0
4	7 430	63.70	121.56	0.00315	38.6
5	8 140	63.75	133.18	0.00394	33.8
6	9 870	64.14	161.49	0.01008	16.0
7	12 850	65.41	210.24	0.03008	7.0
8	14 100	66.68	230.69	0.05008	4.6
9	14 300	67.95	233.97	0.07008	3.3
10	13 830	69.22	226.28	0.09008	2.5
11	12 500	70.49	204.52	0.11008	1.9

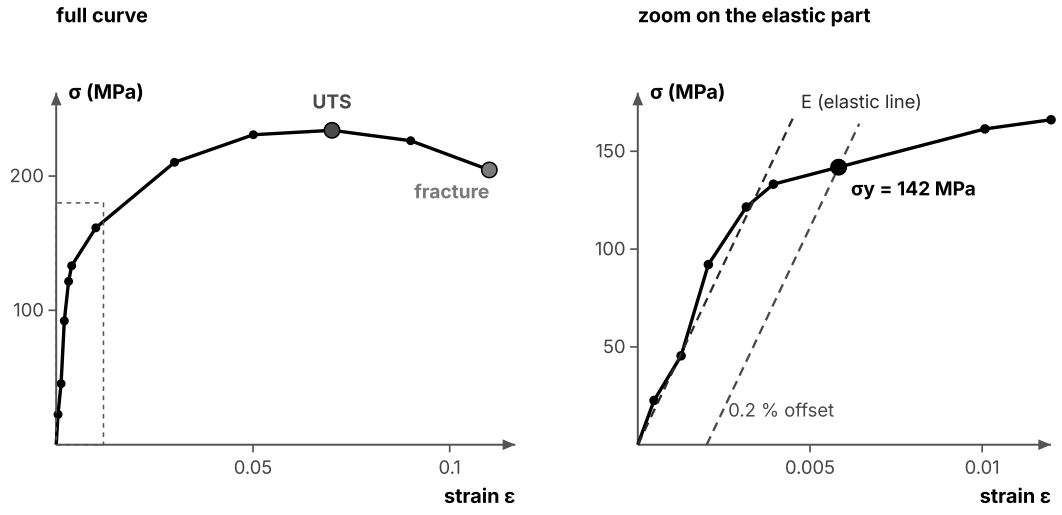


Figure 4. Problem 4: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Modulus of elasticity

The secant modulus of the last column is 34 to 48 GPa for points 1 to 5 and falls to 16 GPa at point 6: the straight part ends after point 5. The scatter between 34 and 48 GPa comes from the resolution of the length: point 1 is an elongation of 0.03 mm, and an error of 0.01 mm is 33 % of it. A fit through the origin on points 1 to 5 weights the points by their strain, so the points with larger elongation, and smaller relative error, count more:

Point	ε_i	σ_i (MPa)	$\sigma_i \varepsilon_i$ (MPa)	ε_i^2
1	0.00047	22.58	0.0107	2.232×10^{-7}
2	0.00126	45.48	0.0573	1.587×10^{-6}
3	0.00205	92.11	0.1886	4.191×10^{-6}
4	0.00315	121.56	0.3829	9.920×10^{-6}
5	0.00394	133.18	0.5243	1.550×10^{-5}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{1.164}{3.1422 \times 10^{-5}} = 37037 \text{ MPa} = \mathbf{37.0 \text{ GPa}}$$

c) Yield strength (0.2 % offset)

With the offset line $E(\varepsilon - 0.002)$ the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ changes between points 5 and 6:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
4	0.00315	121.6	42.6	79.0
5	0.00394	133.2	71.7	61.4
6	0.01008	161.5	299.2	-137.7
7	0.03008	210.2	1040.0	-829.7

$$m = \frac{161.49 - 133.18}{0.01008 - 0.00394} = 4608.6 \text{ MPa} \qquad \varepsilon^* = \frac{133.18 - 4608.6 \times 0.00394 + 0.002 \times 37037}{37037 - 4608.6} = \frac{189.110}{32428.3} = 0.0058316$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 37037 \times (0.0058316 - 0.002) = \mathbf{141.9 \text{ MPa}}$$

d) Tensile strength

The maximum load is $F_{max} = 14\,300$ N (point 9):

$$\sigma_{UTS} = \frac{F_{max}}{A_0} = \frac{14\,300\ \text{N}}{61.12\ \text{mm}^2} = \mathbf{234.0\ MPa}$$

e) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(141.9\ \text{MPa})^2}{2 \times 37037\ \text{MPa}} = \mathbf{0.272\ MPa}$$

f) Working stress

$$\sigma_w = \frac{\sigma_y}{N} = \frac{141.9}{2} = \mathbf{71.0\ MPa}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** N/mm² is MPa, strains have no units, E and U_r are in MPa.
- **Order.** $\sigma_y = 141.9$ MPa $< \sigma_{UTS} = 234.0$ MPa; after the maximum the load falls (point 11 has 12 500 N), the sample necks.
- **Sensitivity to the points chosen for E.** The modulus, and through it σ_y and U_r, depend on how many of the first points are taken:

Points used for E	E (GPa)	σ_y (MPa)	U_r (MPa)	σ_w (MPa)
1 to 3	42.7	139.3	0.227	69.6
1 to 4	40.2	140.4	0.245	70.2
1 to 5	37.0	141.9	0.272	71.0

Between the three choices the yield strength changes by 1.9 % and U_r by 20 % (because it contains $1/E$): the strengths hardly depend on the choice made for E, but the resilience does.

Interpretation

The brass has a tensile strength of 234.0 MPa and a yield strength of about 141.9 MPa, and between the two the strain hardening adds 92 MPa. The large strain at fracture (11 % at the last point, after the maximum load) shows that it is ductile.

THE MODULUS IS MUCH LOWER THAN THE ONE OF BRASS IN THE SLIDE

The table of Session 11 gives $E = 97$ GPa for brass [Session 11 Slide 17]. The data of this problem give 37 to 43 GPa, less than half. The key also gives 44 GPa, so the sheet itself uses these data. The reason is probably that the elongation of the table includes the slack and the compliance of the testing machine in the first points, which is why the elastic part looks softer than the real material. The data are used as they are.

Check against the official solution

Item	Official	Computed	Comment
b. E	≈ 44 GPa	37.0 GPa	differs 16 %: see the sensitivity table; with points 1 to 3 the fit gives 42.7 GPa
c. σ_y	150 MPa	141.9 MPa	differs 5 %
d. σ_{UTS}	234 MPa	234.0 MPa	agrees
e. U_r	0.28 MPa	0.272 MPa	agrees (3 %)
f. σ_w	75 MPa	71.0 MPa	consistent with σ_y

DISCREPANCY WITH THE OFFICIAL SOLUTION: MODULUS AND YIELD STRENGTH

The key gives $E \approx 44$ GPa and $\sigma_y = 150$ MPa. The 0.2 % offset on the data of the table gives σ_y between 139 MPa ($E = 44$ GPa) and 142 MPa ($E = 37.0$ GPa), never 150 MPa. A yield strength of 150 MPa corresponds to the stress where the curve bends clearly (points 5 and 6, 133 to 161 MPa), that is, to a reading by eye of the end of the straight part. In addition, with the key values, $U_r = 150^2/(2 \times 44\,000) = 0.256$ MPa, not 0.28 MPa; the computed resilience, 0.272 MPa, is close to the printed one only by coincidence. The computed values are kept.